

Capstone 3 Proposal

For capstone 3, I want to create an image recognition model that is focused on determining if people are wearing a mask or not. The reason why I believe this is a problem is that after the pandemic, people have become more aware of illness and as a result have taken to wearing masks more frequently. The criteria for success for this project is that given a sample dataset of images consisting of participants who photographed themselves at various angles wearing and not wearing masks, the model will be able to correctly determine if a person is wearing a mask or not at least 70-80 percent of the time. Currently within this space, computer vision is currently being implemented with high quality facial scans from an array of cameras and sensors. However, this is greatly outside the scope of this project which seeks for a rudimentary approach of just being able to determine if there is another layer on top of a person's mouth. For further limitations and constraints on this project, we cannot source any more facial images for the dataset, along we cannot incorporate a varied sensor suite to get higher quality data. Since we are using a singular dataset for this project, there should be no fears of any varied results between different users as well. Now when it comes to stakeholders, the people who are asking for this project are people working in local clinics who do not have much funds available to get sophisticated systems as well as staff who can monitor people compliance with mask wearing. Another client for this project is the open source community who are building on advancing facial recognition software to more accurately determine real faces. Now for this project, this project will be using the MFR2 dataset which is a dataset consisting of 53 unique people wearing

and not wearing a mask totaling 269 different images. For this project deliverable, it will be the end results of this project's computer vision recognition software, the project report and a presentation based on the project's findings.